

On The Total Product Cordial Labeling of $P_n + mK_1$

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Abstract

A total product cordial labeling of a graph $G = (V, E)$ is a mapping $f : (V \cup E) \rightarrow \{0, 1\}$ such that $f(xy) = f(x)f(y)$ where $x, y \in V(G)$, $xy \in E(G)$, and the total number of 0 and 1 are balanced. That is if $v_f(i)$ and $e_f(i)$ denote the set of vertices and edges which are labeled as i for $i = 0, 1$ respectively, then $\left| \left(|v_f(0)| + |e_f(0)| \right) - \left(|v_f(1)| + |e_f(1)| \right) \right| \leq 1$. If there exists a total product cordial labeling on a graph G , then we call G is total product cordial. A total product cordial labeling problem is the problem of determining the graph that is total product cordial. This paper established $G = P_n + mK_1$ is total product cordial when n is odd integer or both n and m are even, and provided an algorithm to check if G is total product cordial when n is even integer and m is odd.

1. Introduction

Cordial labeling which is a weakened version of graceful labeling [2][3][19] and harmonious labeling [9], was first introduced by [4]. Consider $f : (V \cup E) \rightarrow \{0, 1\}$, a binary labeling on graph G such that $f(xy) = |f(x) - f(y)|$ where $x, y \in V(G)$ and $xy \in E(G)$. Let $v_f(i)$ and $e_f(i)$ denote the set of vertices and edges with label i for $i = 0, 1$ respectively, if f satisfies $\left| |v_f(0)| - |v_f(1)| \right| \leq 1$ and $\left| |e_f(0)| - |e_f(1)| \right| \leq 1$, then f is a cordial labeling of G . A graph G is cordial if there is a cordial labeling of G . The cordial labeling problem is the problem of

determining whether the graph is cordial graphs which has been well studied [1][5][8][10][11][13-18][20-22]. Later on, [6] defined total sequential cordial labeling which instead of asking $\left| |v_f(0)| - |v_f(1)| \right| \leq 1$ and $\left| |e_f(0)| - |e_f(1)| \right| \leq 1$, it asked for $\left| \left(|v_f(0)| + |e_f(0)| \right) - \left(|v_f(1)| + |e_f(1)| \right) \right| \leq 1$, and they proved that every cordial graph is total sequential cordial. There are other versions of cordial related problems such as uniform cordial [7], total magic cordial [6], k -cordial [12], etc. In 2004, [23] defined product cordial labeling, in which the edge gets the label as the product of two end vertices instead of the difference. In 2006, [24] discussed total product cordial of graphs obtained as the join of two graphs. The total product cordial labeling is defined as a binary labeling $f : (V \cup E) \rightarrow \{0, 1\}$ such that $f(xy) = f(x)f(y)$ where $x, y \in V(G)$, $xy \in E(G)$, and the total number of 0 and 1 are balanced. That is if $v_f(i)$ and $e_f(i)$ denote the set of vertices and edges which are labeled as i for $i = 0, 1$ respectively, then $\left| \left(|v_f(0)| + |e_f(0)| \right) - \left(|v_f(1)| + |e_f(1)| \right) \right| \leq 1$. [24] proved that $P_n + K_1$ and $P_n + 2K_1$ are total product cordial. In this paper, we showed a stronger result for general $P_n + mK_1$ to be total product cordial.

2. Main result

Let $f : (V \cup E) \rightarrow \{0, 1\}$ be a 0-1 labeling on graph G . Let $v_f(0)$ and $v_f(1)$ denote the set of vertices which are labeled as 0 or 1 respectively. Similarly, let $e_f(0)$ and $e_f(1)$ denote the set of

edges which are labeled as 0 or 1 respectively. Consider $G = P_n + mK_1$, let the vertices of P_n be denoted as $v_1, v_2, \dots, v_n$ and the vertices of mK_1 be denoted as $u_1, u_2, \dots, u_m$ for $m, n \geq 1$. So $V(G) = \{v_i : 1 \leq i \leq n\} \cup \{u_j : 1 \leq j \leq m\}$ and $E(G) = E(P_n) \cup \{u_i v_j : 1 \leq i \leq m, 1 \leq j \leq n\}$. Then we know that $|V(G)| = |V(P_n)| + |V(mK_1)| = n + m$, and $|E(G)| = |E(P_n)| + |\{u_i v_j : 1 \leq i \leq m, 1 \leq j \leq n\}| = (n-1) + nm$. Hence if there is a total product cordial labeling f on G , $|v_f(0)| + |e_f(0)|$ must be no more than $\lceil (nm + m + 2n - 1)/2 \rceil$.

Lemma 1: If $G = P_n + mK_1$ is total product cordial then $|\{v \mid f(v) = 0, v \in P_n\}| \leq n/2$ for any total product cordial labeling f of G .

Proof: Let $|\{v \mid f(v) = 0, v \in P_n\}| = t$. Assume to the contrary that $t \geq \lfloor n/2 \rfloor + 1$. Let f be a total product cordial labeling on G . If there are t vertices in P_n which is labeled 0, then there must be at least t edges in P_n which is labeled 0, and tm edges between P_n and mK_1 which are labeled 0. Hence we have $|v_f(0)| + |e_f(0)| \geq 2t + tm = t(m+2) \geq (n+1)(m+2)/2 > \lceil (2n + nm + m - 1)/2 \rceil$ which is a contradiction. ■

For the following theorems, according to different condition on n and m , we define a binary labeling on graph G such that $f(xy) = |f(x) - f(y)|$ where $x, y \in V(G)$ and $xy \in E(G)$; show the labeling has balanced 0 and 1, and therefore is a total product cordial labeling of G which implies G is total product cordial.

Theorem 1: Let $G = P_n + mK_1$ for some odd n . Then G is total product cordial.

Proof: Let $n = 2k + 1$, then $|V(G)| + |E(G)| = nm + m + 2n - 1 = m(2k + 2) + 4k + 1$. Suppose there is a total product cordial labeling f on G , then we must have $|v_f(0)| + |e_f(0)|$ be either $m(k+1) + 2k$ or $m(k+1) + 2k + 1$. Notice that if

$f(v_i) = 0$, and $f(v_{i-1}) = f(v_{i+1}) = 1$, then we have $f(v_{i-1}v_i) = f(v_i v_{i+1}) = f(v_i) = 0$ and $f(u_i v_j) = 0$ for $\{v_i u_j : 1 \leq j \leq m\}$. That is whenever we label a vertex v_i in P_n to be 0 that will force $m+2$ edges to be labeled as 0, hence that should be counted as $m+3$ zeros. Suppose there are a this kind of vertices. Similarly, if $f(v_i) = 0$ and only one of its neighbor in P_n is labeled as 1, that should be counted as $m+2$ zeros. Suppose there are b this kind of vertices and let $a + b = k$, then there are $a(m+3) + b(m+2)$ vertices and edges labeled as 0. Now consider the vertices in mK_1 . When we label a vertex u_j to be 0, it will cause $f(u_j v_i) = 0$ for those $f(v_i) = 1$. Since there are $k+1$ those edges, that will give us total of $k+2$ zero labels. Suppose there are c this kind of vertices, that will give us total of $a(m+3) + b(m+2) + c(k+2)$ vertices and edges to be labeled as 0. If f is a total product cordial labeling of G , there should be either $m(k+1) + 2k$ or $m(k+1) + 2k + 1$ vertices and edges to be labeled as 0. That is $a + c(k+2) = m$ or $m+1$. If $m \bmod (k+2) \leq k$ then we can take $c = \lfloor m/(k+2) \rfloor$ and $a = m \bmod (k+2)$. If $m \bmod (k+2) = k+1$, say $m = x(k+2) + (k+1)$, then we can take $c = x+1$, and $a = 0$ which will make $a + c(k+2) = m+1$. Therefore, for any non-negative integers m and k , there is a total product cordial labeling of G which makes $|v_f(0)| + |e_f(0)| = m(k+1) + 2k$ or $m(k+1) + 2k + 1$. Hence G is total product cordial When n is odd. ■

Theorem 2: If n and m are even integer, $P_n + mK_1$ is total product cordial.

Proof: Let $n = 2k$ and $m = 2t$, $k, t \geq 1$. Then we have $|V(G)| + |E(G)| = |V(P_n)| + |V(mK_1)| = n + m$, and $|E(G)| = |E(P_n)| + |\{u_i v_j : 1 \leq i \leq m, 1 \leq j \leq n\}| = (n-1) + nm$, which implies $|V(G)| + |E(G)| = nm + m + 2n - 1 = 2mk + 2t + 4k - 1$. Suppose there is a total product cordial labeling f on G , then

we must have $|v_f(0)| + |e_f(0)| = mk + t + 2k$ or $mk + t + 2k - 1$. Notice that if $f(v_i) = 0$, and $f(v_{i-1}) = f(v_{i+1}) = 1$ then we have $f(v_{i-1}v_i) = f(v_iv_{i+1}) = f(v_i) = 0$ and $f(v_iu_j) = 0$ for $\{v_iu_j : 1 \leq j \leq m\}$. That is whenever we label a vertex v_i in P_n to be 0 that will force $m+2$ edges to be labeled as 0, hence that should be counted as $m+3$ zeros. Suppose there are a this kind of vertices. Since there are $2k$ vertices, we know that $a \leq k-1$. Similarly, if $f(v_i) = 0$ and only one of its neighbor in P_n is labeled as 1, that should be counted as $m+2$ zeros. Suppose there are b this kind of vertices and let $a+b=k$, then there are $a(m+3)+b(m+2)$ vertices and edges labeled as 0. Now consider the vertices in mK_1 . When we label a vertex u_j to be 0, it will cause $f(u_jv_i) = 0$ for those $f(v_i) = 1$. Since there are k those edges, that will give us total of $k+1$ zero labels. Suppose there are c this kind of vertices, that will give us total of $a(m+3)+b(m+2)+c(k+1)$ vertices and edges to be labeled as 0. If f is a total product cordial labeling of G , there should be either $mk+t+2k$ or $mk+t+2k-1$ vertices and edges to be labeled as 0. That is $a+c(k+1)=t$ or $t-1$. If $t \bmod (k+1) < k$ then we can take $c = \lfloor t/(k+1) \rfloor$ and $a = t \bmod (k+1)$. If $t \bmod (k+1) = k$, say $t = x(k+1) + k$, then we can take $c = x, a = k-1$ which will make $a+c(k+1) = t-1$. Therefore, for any non-negative integers m, k , there is a total product cordial labeling of G which makes $|v_f(0)| + |e_f(0)| = mk + t + 2k$ or $mk + t + 2k - 1$. Hence G is total product cordial. ■

Theorem 3: Let $G = P_n + mK_1$, for $n = 2k$ and $m = 2t+1$, $k, t \geq 1$. Then G is total product cordial if $0 \leq [(2i+1)t + 3i] \bmod (k+1+i) \leq (k-i)$ for $i = 1$ to k or $0 \leq t \bmod (k+1) < k$.

Proof: Let $n = 2k$ and $m = 2t+1$, $k, t \geq 1$. Then we have $|V(G)| + |E(G)| = nm + m + 2n - 1$

$= 2mk + 2t + 4k$. Suppose there is a total product cordial labeling f on G , then we must have $|v_f(0)| + |e_f(0)| = mk + t + 2k$. Notice that if $f(v_i) = 0$, and $f(v_{i-1}) = f(v_{i+1}) = 1$ then we have $f(v_{i-1}v_i) = f(v_iv_{i+1}) = f(v_i) = 0$ and $f(v_iu_j) = 0$ for $\{v_iu_j : 1 \leq j \leq m\}$. That is whenever we label a vertex v_i in P_n to be 0 that will force $m+2$ edges to be labeled as 0, hence that should be counted as $m+3$ zeros. Suppose there are a this kind of vertices. Since there are $2k$ vertices, we know that $a \leq k-1$. Similarly, if $f(v_i) = 0$ and only one of its neighbor in P_n is labeled as 1, that should be counted as $m+2$ zeros. Suppose there are b this kind of vertices and let $a+b=k$, then there are $a(m+3)+b(m+2)$ vertices and edges labeled as 0. Now consider the vertices in mK_1 . When we label a vertex u_j to be 0, it will cause $f(u_jv_i) = 0$ for those $f(v_i) = 1$. Since there are k those edges, that will give us total of $k+1$ zero labels. Suppose there are c this kind of vertices, that will give us total of $a(m+3)+b(m+2)+c(k+1)$ vertices and edges to be labeled as 0. If f is a total product cordial labeling of G , there should be $mk+t+2k$ vertices and edges to be labeled as 0. That is $a+c(k+1)=t$. If $t \bmod (k+1) < k$ then we can take $c = \lfloor t/(k+1) \rfloor$ and $a = t \bmod (k+1)$. If $t \bmod (k+1) = k$, we consider $0 \leq a+b \leq k-1$. Let $a+b = k-i$, $1 \leq i \leq k$. We have $a(m+3)+b(m+2)$ vertices and edges labeled as 0. Now consider the vertices in mK_1 . When we label a vertex u_j to be 0, it will cause $f(u_jv_i) = 0$ for those $f(v_i) = 1$. Since there are $k+i$ those edges, that will give us total of $k+1+i$ zero labels. Suppose there are c this kind of vertices, that will give us total of $a(m+3)+b(m+2)+c(k+1+i)$ vertices and edges to be labeled as 0. If f is a total product cordial labeling of G , there should be $mk+t+2k$ vertices and edges to be labeled as 0.

That is $a + c(k+1+i) = [(2i+1)t + 3i]$. If $0 \leq [(2i+1)t + 3i] \bmod (k+1+i) \leq (k-i)$ for $i=1$ to k , say $t = x(k+1+i) + \{[(2i+1)t + 3i] \bmod (k+1+i)\}$, then we can take $c = x, a = [(2i+1)t + 3i] \bmod (k+1+i)$ which will make $a + c(k+1+i) = [(2i+1)t + 3i]$. Therefore, for any non-negative integers m, k , there is a total product cordial labeling of G which makes $|v_f(0)| + |e_f(0)| = mk + t + 2k$. Hence G is total product cordial. ■

3. Conclusion

Cordial labeling of graphs has been discussed for more than twenty years. There are several different aspect of the cordial labeling. This paper deals with the total product cordial, in which the label of the edge depends on the product of two end vertices and the total number of labels 0 and 1 are balanced. This paper proved the graph $G = P_n + mK_1$ is total product cordial when n is odd or both n and m are even. For the case then n is even and m is odd, this paper provided an algorithm to determine if G is total product cordial.

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